

## Review

# A Review of Digital Twin Technology for Electric and Autonomous Vehicles

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**Abstract:** In the era of technological transformation, mobility and transportation systems are becoming more intelligent and greener. Thanks to powerful technologies and tools, electric and autonomous vehicles are spreading worldwide, substituting internal combustion engine vehicles and revolutionizing the way to drive. In this context, this paper is an extended version of the paper “Digital Twin in Intelligent Transportation Systems: a Review published in 2020 7th International Conference on Control, Decision and Information Technologies (CoDIT)”. The aim of this paper is to provide a comprehensive review of the literature from the last five years on the use of digital twin (DT) technology for Intelligent Transportation Systems (ITSs), focusing on electric and autonomous vehicles. In particular, with respect to the previous work, the focus has been expanded to include DT integration with other cutting-edge technologies, such as the Internet of Things (IoT), Big Data, artificial intelligence (AI), machine learning (ML), and 5G for ITS. Moreover, this paper presents a broad perspective on challenges in EV applications, including tracking, monitoring, battery and charge management, connectivity, security, and privacy. In addition, this paper discusses how DT can be used to effectively address the current issues in electric vehicle services, such as tracking, monitoring, battery and charge management, connectivity, security, and privacy.

**Keywords:** digital twin; intelligent transportation system; electrical vehicles; autonomous vehicles; 5G; IoT; Big Data analytics



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## 1. Introduction

In the era of the fourth industrial revolution, the manufacturing field of the transportation system is being transformed by the electric vehicle (EV) industry. Today, most EV features and services are realized through smart technology. Conventional vehicles with internal combustion engines significantly contribute to the consumption of fossil fuels and the emission of greenhouse gases, such as carbon oxides and hydrocarbons. To overcome this issue, EVs have been improved over the past few years [1]. Many industrial fields adopt the Internet of Things (IoT) technologies to enhance the electromobility industry and make this transformation smarter. Vehicles are becoming smart objects by using sensors that form the basis for IoT networks. In turn, the amount of data these sensors provide will lead to a qualitative shift in the concept of EV management systems. These sensors will cover all vehicle parts to monitor all movements and changes during charging and engine monitoring, as well as internal components. The generated data from the sensors need to be collected, synchronized, analyzed, and processed to improve the EV service quality and assist the EV management system with decision-making. Despite the significant development of these technologies, there are still difficulties in using physical sensors; this has led many researchers to introduce the concept of virtual sensors (VSs) for electromobility [2]. The VS system can analyze, predict, and estimate the vehicle's behavior, the battery state of charge (SoC), and the availability of charge points.

Improving the EV user's experience relies on three essential components: the physical entities in the real world, the virtual models, and the data driven by these models. Integrating these components requires a simulation framework to simulate large-scale traffic scenarios [3]. The distribution of charging points, EV volume, and all dynamic operations in the EV network should be managed effectively and safely. To achieve this goal, simulation platforms can simulate the components of the EV network and their interactions. The simulation of operations could help us understand the nature of the physical product in each stage and collect information about the product's characteristics, which can be helpful in the development process. Most simulation platforms support the digital twin (DT) concept, which can simulate the real-world entity in the industrial environment. The DT concept is known as a virtual replica of a real-world object that can provide the ability to study the development of physical objects in a digital situation/environment. DT was considered one of the world's ten most strategic innovations in 2019, providing autonomous objects (e.g., self-driving cars) and immersive technologies, such as virtual reality (VR), augmented reality (AR), and quantum computing [4]. The main idea of this technology is to replicate the physical object's behavior in a virtual environment that can produce the same output as the actual physical object.

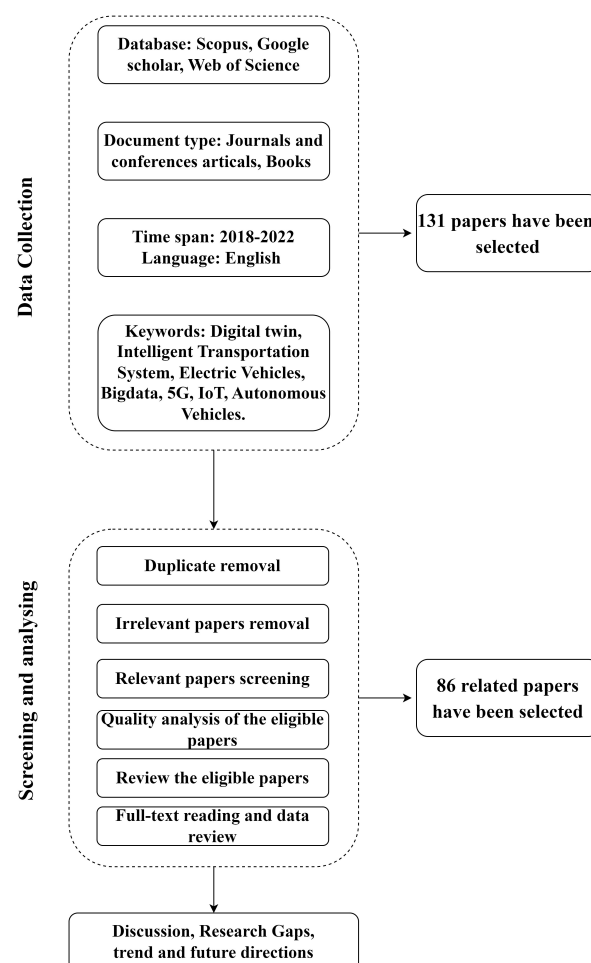
Adopting the DT concept can reinforce the development of the industry and the academic section. Digital data can improve an engineering system's intelligence concerning the analytical evaluation, extrapolative diagnosis, and performance optimization. The results of the analysis can be used to make the product or process run better in the physical environment [5]. Academically, the DT concept was introduced in 2002 by Grieves et al. [6] at a special summit on product life-cycle management at the University of Michigan Lurie Engineering Center. The first adoption of this technology was by Tuegelet al. [7], who presented a digital framework to reproduce the structural behavior of an aircraft. Indeed, DT technology is widely used in multiple industrial sectors to facilitate maintenance operations and predict failures, allowing machines to interact with each other and humans. In particular, DTs are used in a wide range of applications, including transportation, manufacturing, medicine, business, education, and more. Integrated machine-driven, electrical, and computer software systems can be simulated in the virtual workspace through DT technology [8]. One important example of such systems involves EVs, where new technology is needed to continuously optimize vehicle performance. DT technology can be an innovative solution for EV optimization. Many features of an electric EV can be computerized to increase its efficiency, performance, and smartness.

Integrating Big Data analytics, IoT, and artificial intelligence (AI) technologies with DT leads to new significance, prospects, and challenges. Furthermore, an intelligent DT model can only be created using advanced AI technologies applied to the data [9]. It can address significant challenges in manufacturing, such as improving process stability, fault diagnosis, reducing downtime, and optimizing logistics processes [10]. AI can further enhance DT technology by using analytical models to process raw data into valuable digital forms. In this context, machine learning (ML) algorithms and technologies are currently used in EVs. In particular, ML algorithms can be used effectively if combined with predictive testing tools and DT technology. The importance of DT is also proved in the security and monitoring systems. Lu et al. [11] presented a DT-enabled anomaly detection system based on industry foundation classes for asset-monitoring solutions. The proposed framework was evaluated using a case study to control the Heating, ventilation, and air conditioning system, and the system efficiently contributes to monitoring building assets.

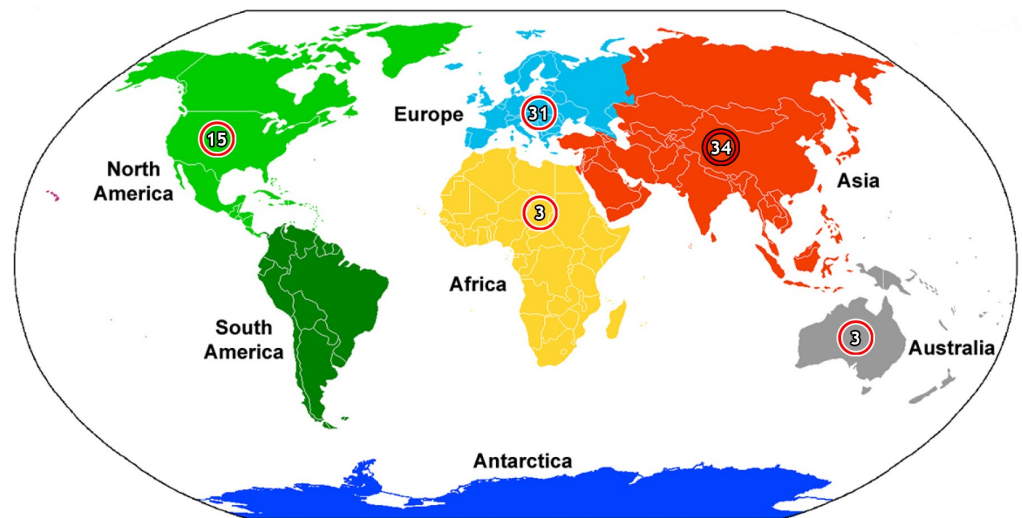
There is a lack of literature reviews that explore the use of DT in Intelligent Transportation Systems (ITSs), especially for EVs and autonomous vehicles (AVs). Due to the rapid development of the smart mobility industry, there is an urgent need to study and analyze the challenges and issues that the new generation of transportation will produce and how these can be addressed [12]. In response to the growing adoption of DT technology in ITS, this survey investigates recent literature, particularly emphasizing the application of DT in electromobility and self-driving systems. By exploring the integration of emerging

technologies, communication tools, and DT concepts, our findings reveal the potential of DT technology to address challenges, such as cost-effectiveness, reliability, visualization, charging time, and intrusion detection in electric and autonomous vehicle networks. Furthermore, this survey highlights the role of data analytics and machine learning techniques in securing the ITS network and improving overall efficiency. We searched multiple databases, including Scopus, Google Scholar, and Web of Science, to identify relevant studies published in the last five years. In addition, we considered additional databases and information sources, such as ResearchGate.

The provided review was carried out systematically, as shown in Figure 1, considering specific domains within smart manufacturing and transportation in which DT technology is applied, in combination with IoT, ML, AI, and 5G technologies. First, we selected a set of papers from the publication databases by considering the following keywords: DT, ITS, EVs, Bigdata, 5G, IoT, and AVs. From such paper sets, we singled out a subset of articles on the basis of their relevance and quality. Then, the paper discusses the current issues in EV services, such as tracking, monitoring, battery management systems (BMSs), connectivity, privacy, and security, and how they can be addressed effectively through DT technology. We highlighted the next EV revolution, i.e., electric AVs, and discussed the importance of data analytics roles in applying the DT concept in such a context. In Figure 2, a geographical distribution map displays the dispersion of the research articles examined in this review paper. The map visually represents the number of papers originating from each continent, providing a clear overview of the global research landscape in this field. This illustration allows readers to easily grasp the contributions on DT and ITS made by researchers from different parts of the world, showing the regions with the most research output.



**Figure 1.** Systematic review flow chart.



**Figure 2.** Visualization of the geographical distribution of the analyzed articles.

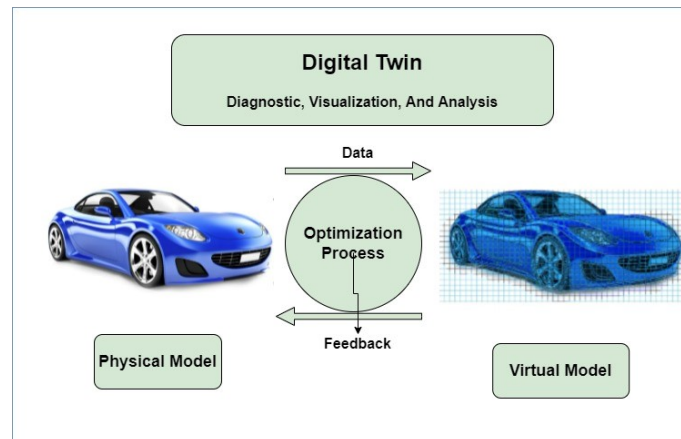
This paper is organized as follows. Section 2 discusses the application of DT to transportation systems. Section 3 analyzes the key technologies adopted in DT for electromobility. Section 4 presents the current electromobility issues. Section 5 discusses the DT solutions for the identified challenges, and Section 6 provides the final outcomes of this review study. Finally, Section 7 presents the concluding remarks. The table of abbreviations presents the names reported at the end of this paper.

## 2. Digital Twin in Transportation Systems

With the advancements made in Big Data, IoT, and AI, a new generation of information, technology, geographic data, and global positioning data, is to be handled. Combining these technologies with DT technology is critical in digital wave trends and takes the lead in transportation applications for planning, maintenance, security, and other aspects.

The DT concept has the potential to improve the transportation sector by providing a digital identity, synchronized visualization, and virtual and real interactions (see Figure 3). DT technology utilizes intelligent technical advantages, such as controlling traffic perception, road warnings, and emergency responses. Furthermore, it can provide transportation solutions and new paths, such as intelligent driving, which will increase efficiency and safety and allow convenient traffic management. In this context, Wang et al. [13] introduce a DT framework for connected vehicles using an advanced driver assistance system (ADAS). They used vehicle-to-cloud communication to calculate the advisory speed based on the information that could be collected from the sensors on the vehicles. The proposed model helps the driver to control the speed intelligently. Another example of using DT in the cloud was presented by Alam and Saddik [14], who developed a DT model for the cloud-based cyber-physical system (C2PS). They described the key properties of the C2PS and introduced a telematics-based prototype driving assistance application for the vehicular C2PS.

In this context, a review of DT technology with smart EVs was conducted by Bhatti et al. [15]. The review divided the smart vehicle systems into different categories: autonomous navigation control, advanced driver assistance systems, vehicle power electronics, vehicle health monitoring, BMS, and electric power drive systems. As a result of this research, smart EVs and DT technology were investigated theoretically to see what impacts their integration could have in the near future.



**Figure 3.** Digital twin model in transportation.

Due to the amount of data generated by transportation systems, machine learning (ML) and deep learning (DL) technologies are employed to create an ITS. The application of intelligence in the transportation field is increasing rapidly and improving the performances of these systems has become the research focus. The existing transportation systems can be controlled, analyzed, and operated using the DT concept. Using ML and DL in the DT concept can provide real-time data and effective services to the service provider and end-user [9]. Moreover, the ITS can effectively optimize and coordinate traffic conditions based on DL and DT technologies by monitoring the flow of people, traffic, and roads. It can also maximize the duration of traffic lights and find the signal light scheme with the shortest transit time. For example, Zhihan et al. [16] proposed a DL algorithm to solve the security problems of the ITS. The proposed model assured a response time to emergency alerts and increased prediction accuracy. Moreover, vehicles will travel faster because they can better adapt to the road environment, transmit data faster, and develop routes by considering traffic patterns. DT and AI technologies are utilized in transportation via traffic management, prediction, and congestion avoidance. Kumar et al. [17] introduced an ITS that uses ML, fog/edge analytics, data lakes, DT, and blockchain. The authors used cameras to collect environmental information and then ran edge analytics on the collected data. The DT generated the virtual car model to simulate the real-world scenario. This work used ML and DL algorithms to predict driver intentions. By creating a virtual vehicle model, non-autonomous drivers were able to make better decisions depending on the current traffic scenario and the intents of other drivers.

The DT can also improve the driving experience for travelers by reducing and redistributing wait times at intersections. Dasgupta et al. [18] worked on a DT approach for adaptive traffic signal controls to improve the user's driving experience. They developed DTs to emulate vehicles that were close to the intersection and vehicle wait times at the immediate upstream intersection. The proposed model can balance wait times across a signalized network to improve the driving experience in congested areas, and it can be scalable on the city-wide network. While the data analytics in the DT concept is still being developed, Aslani et al. [19] developed a real-time DT simulation model that can provide performance measures. The study also demonstrated the data scarcity required for real-time applications that rely on high real-time frequency-connected corridor data streams. In summary, the DT uses all of the gathered data and accurately captures city signs to achieve new insights into urban traffic on different sides, such as the road supply and traffic demand, optimizing the road network structure through traffic simulation, and improving overall traffic efficiency in the city. In addition, using DT in transportation can improve the decision-making execution, safety, and stability of vehicle driving and accelerate intelligent and safe driving.



### 2.1. Digital Twin in Electromobility

Many applications of electromobility have been involved in research activities, leading to the phenomenon known as smart electromobility. The rapid growth of smart control systems has led to several developments in this industry. The data generated through this growth are the key factors in improving the smart mobility sector. The DT's strength lies in collecting and visualizing data and conducting statistics in which advanced analysis tools are used to improve manufacturing processes and help in decision-making.

To charge an EV, it is commonly vital to physically attach/connect the EV plug to a charger located in a household or a public place through a charging cable. As EVs and self-driving cars become more prevalent and automated driving becomes more common, physical charging may become unmanageable, and automatic charging systems should be implemented. Generally, there are two options for automatic charging: parking the vehicle in an accurate position so that the vehicle's charging connector automatically fits the charging cable of an available charger or using wireless charging. Shikata et al. [20] introduced a DT vehicle simulation technique, focusing on two factors: (1) power consumption and (2) ride comfort. The simulated environment contains a vehicle model for interpreting the physical performance of the vehicle. An electronic control unit (ECU) has also been simulated as a prototype for regulating the simulated environment. They also developed an automatic charging system for EVs to charge the vehicle automatically after parking in an accurate position.

BMS in electromobility is also essential concerning battery life, safety, and reliability. It relies on different types of sensors and actuators on the EV to provide real-time battery performance. Using an IoT platform to build a DT for BMS in the cloud boosts the robustness of the BMS. Wang et al. [21] reviewed the solutions for BMS issues based on DT, such as the problems related to real-time estimation, dynamic charging control, and dynamic equalization control in a smart BMS. Another important application of the electric vehicle is the ADAS, built to enhance the driver's experience and passenger and pedestrian safety by decreasing vehicle accidents and alerting drivers of possible dangers. Liu et al. [22] introduced a new vision-cloud data fusion approach to enhance the performances of visual guidance systems by leveraging DT technology and cloud servers. This work is one of the most effective studies to visualize the DT cloud data and support the ADAS or the drivers' decision-making.

### 2.2. Digital Twin Networks in Smart Transportation

A digital twin network (DTN) is the natural evolution of the development of DT technology in the modern era. The DT of any physical object is the first cell of the DTN; thus, we can define the DTN as a set of virtual digital representations of different groups of physical objects connected by a high-speed communication medium to configure an integrated virtual system. The data exchange between the virtual model and physical object in the DT is done in a one-to-one unidirectional way. The operational changes in the physical object will directly affect the virtual model but not the opposite.

On the other hand, the DTN allows for the comprehensive data exchange between DTs and physical assets in a multidirectional manner [23]. Recently, transportation has encountered issues that have increased with the development of urban cities, such as traffic congestion and accidents. In this context, the DTN can provide a better solution for such a complex environment and help to optimize the entire transportation system. The DTN also offers innovative transportation services, such as traffic information reporting, vehicle security, and data sharing. To keep pace with the rapid progress in the electric mobility sector, we need to use and integrate DTN technology with EV networks in smart cities, whether involving autonomous or non-autonomous vehicles, which will provide high possibilities for managing and improving transportation network systems at the city level as well as at a broader level.

We consider the DTN architecture shown in Figure 4, composed of three layers: physical, network, and virtual. The physical layer consists of EVs, charging stations, roads,

and facilities. These entities are connected to the network layer through sensors that transmit data concerning the vehicle positions and velocities, the road traffic, and the charging station status. The network layer receives information from the physical layer by communication services provided through 5G or WIFI technologies. Moreover, this layer sends information and data to the virtual layer composed of a network of DTs and servers. At the virtual level, the DTs are connected to collaborate in executing the simulation and computation tasks based on new enabling technologies devoted to decision-making, analysis, and maintenance issues (such as AI, AR, and ML). In this context, Dai et al. [24] proposed a new DTN model to build the network topology and integrate it with the IoT network. The adopted system has significantly solved many problems, such as computation offloading as well as resource allocation problems.

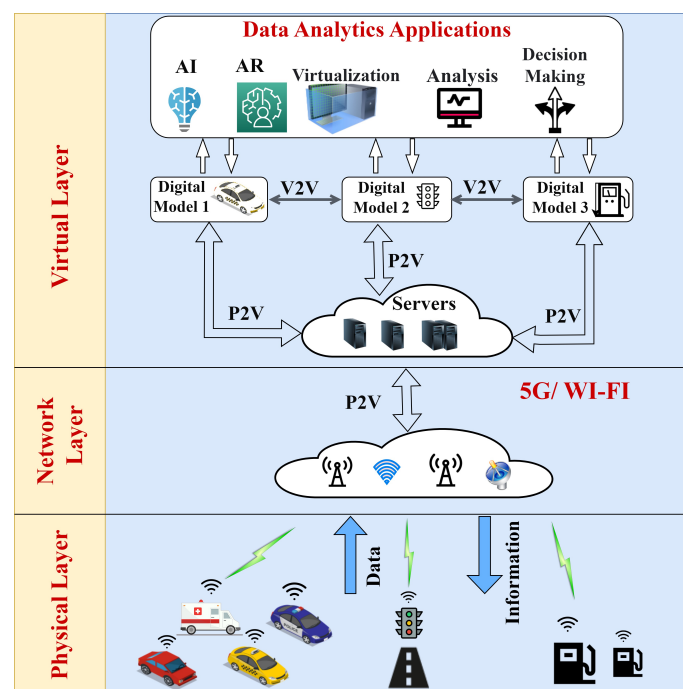


Figure 4. DTN architecture for electromobility.

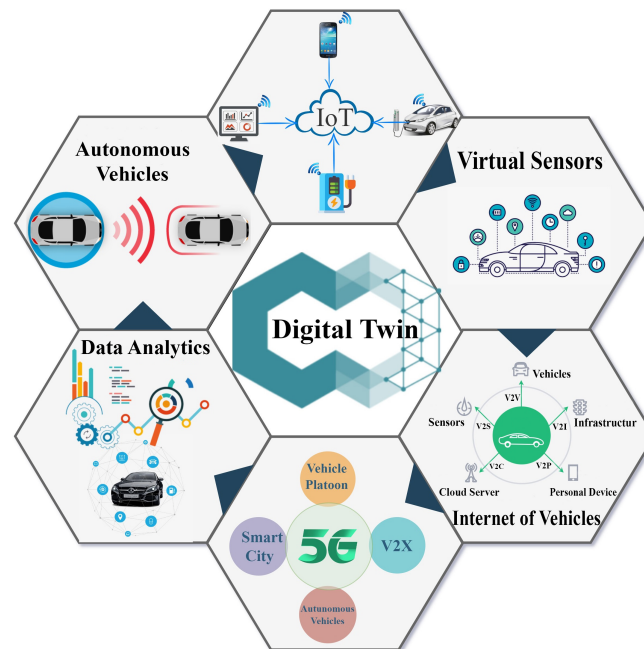
### 3. Digital Twin Technologies for Electromobility

DT applications are increasingly being used in electromobility and are combined with enabling technologies, such as IoT, 5G, Big Data, ML, virtual systems, and advanced communication interfaces. Critical functionalities, such as real-time monitoring, predictive analysis, or cloud computing, may be impacted. However, the main concept and basic architecture of DTs are common across all these technologies. Figure 5 shows the key technologies to be used with the DT for electromobility, such as the Internet of Things, virtual sensors, 5G, data analytics, and autonomous vehicles. We will discuss the main contributions of the DT in detail in this section and how DT technology will provide services to heterogeneous fields in different communication networks.

#### 3.1. Digital Twin and Internet of Things

Recently, IoT technology has been used in smart and electric mobility. In the electric mobility revolution, DT technology applications are facilitated by advanced data analytics and IoT. Digital and physical interactions are changed by integrating DT and IoT platforms. The IoT enables connections and intelligent access to physical devices, and the DT can handle challenges related to integrating IoT and data analytics, which facilitate rapid real-time analyses and decisions. In electromobility, the IoT has established a wide platform with connected vehicles that can send data from physical devices to the cloud or local

servers. The role of the DT lies in dealing with this information, simulating resources by creating DT models, establishing virtual connections, and integrating with artificial intelligence. Therefore, DT technology is a substantial technology used to innovatively improve performance in electromobility and advances monitoring, analytics, and predictive capabilities [25].



**Figure 5.** Key DT technologies for electromobility.

Zhao et al. [26] introduced IoT-based and DT-based models that enable tracking solutions for safety management. The proposed framework allows for a real-time safety tracking mechanism for detecting stationary behavior and self-learning genetic positions to recognize abnormal conditions and obtain accurate locations. The combination of IoT and DT technologies can help operators and stakeholders in taking the necessary technological improvements for making electromobility smarter, by connecting DTs of smart vehicles, simulating and managing the EV fleet data network for value-added services that improve the driving and charging experiences of users, and generating benefits for the entire sector. Being able to simulate the connection of EVs through IoT can, for instance, allow for optimally scheduling the recharging of such vehicles by prioritizing the charging operations based on real-time states of charge, available infrastructure, and/or user preferences. Finally, DT technology facilitates smart functionalities by leveraging data gathered consistently from IoT devices. Moreover, DT enables features such as predictive maintenance, network analysis, energy efficiency optimization, streamlined resource distribution, and real-time network supervision. Additionally, seamless interactions between various networking devices are possible, as their digital counterparts are platform-agnostic and can be managed using standardized methods without concern for the specific technical details of individual devices [27].

### 3.2. Digital Twin and Virtual Sensors

EVs use many environmental sensors to perceive and act according to their perceptions. This section illustrates how logical entities, called virtual sensors (VSs), can be used in a DT framework as the virtual twins of physical sensors. VSs can support new services for smart transportation, addressing issues related to battery charging and route planning for EV drivers based on parameter estimation/prediction. In particular, VSs derive new data from existing information generated by physical sensors and utilize a data processing algorithm to process the data input and produce the required output [28].



For example, Roccotelli et al. [2] introduced and designed new virtual sensors to enhance the EV charging experience. The proposed model provides a smart charging service that allows drivers to find the best charging points for their vehicles. Another example is given by Gruosso et al. [29], who proposed a methodology for estimating the state of charge in EVs. The method relies on VSs and other measurements available in the vehicle, such as speed, acceleration pedal position, and battery voltage. VSs also play an essential role in enhancing user experiences and optimizing EV services, which could support the growth of the EV market.

In [30], Fanti et al. developed a new EV service to improve user experiences while preparing for a trip. They designed three VSs that help the driver predict the cost and required energy for the journey. The proposed VSs can estimate the energy demand based on historical data from past trips. Some relationships between the sensors can be determined over time by the virtualization platform, for example, by utilizing and exploiting ML technologies to improve the functioning of the sensors. Therefore, combining the technology of VSs with the EV simulation model can provide a way to solve complicated issues, such as battery management, vehicle energy management, and vehicle control.

### 3.3. Digital Twin for Internet of Vehicles

By guaranteeing greenhouse gas reduction and fuel efficiency, EVs are attracting a greater share of the private automobile market [31]. Nevertheless, the charging problem is still a barrier to the EV business growth with the present battery technology. Thus, it is essential to have a wide charging infrastructure area that holds fast charging poles, battery swapping stations, and individual charging points for faster EV battery charging. In this scenario, the EV model is simulated with the DT to accurately replicate the EV in the real world. Introducing a DT model makes it easy to simulate mobility behaviors and interactions to study the efficiency of the charging pole and EVs from the demand-side and supply-side.

DT for IoV creates virtual representations of vehicles and traffic systems, connecting the virtual world and physical world spaces. This allows for real-time vehicle and traffic system performance monitoring, improved situational awareness, feasibility forecasting, and decision-making through comprehensive, multidimensional modeling. Continuous, real-time interactions between the virtual and physical realms are necessary for accurate simulations [32].

One example of the potential of DT for IoV is found in [33]. In this work, a simulation platform based on DT models is proposed to replicate and simulate charging and discharging processes of large-scale EV fleets in different kinds of dynamic scenarios. To replicate the realistic mobility of EVs, the operational motion parameters, such as steering angle, orientation angle, and acceleration, are integrated into the mobility model to simulate the realistic trajectories. A tracking method of the dynamic position and orientation is proposed to synthesize the reasonable trajectory of EVs in day-to-day traffic conditions. In this framework, in which the key elements of the real world are considered, the digital twins of EVs and charging points behave and interact with each other as real entities; the simulation results can be used to evaluate schemes for the deployment and management of EVs and charging infrastructures. In addition, the DT simulation platform can be used to verify the design of the deployment of the charging infrastructure and related impacts on the smart grid [34]. The advancements in this area can solve the problem of managing and exploiting real-time traffic data. The large amounts of traffic data available could help in creating a DT that creates a virtual representation of physical vehicles via various communication means.

### 3.4. Digital Twin and 5G Networks

The 5G network supports a wide range of applications in different industries. The enhancement of the 5G network communication has a significant impact on Industry 4.0. Such industries are smart cities, military applications, healthcare systems, and transportation

using IoT. The 5G network makes rapid changes in wireless communications and improves the wireless network performance by increasing capacity, improving reliability, lowering latency, and increasing the network speed [35]. The previous cellular technologies depend on fixed infrastructure, while 5G networks significantly enhance the use of small cell sites and mobile cell sites, increasing network access in congested areas. The 5G network has various applications and dynamically changes latency, bandwidth, and reliability requirements. These requirements have a high impact on the deployment of the 5G network in EVs.

The 5G network is largely used in EVs for communication between the EV components. The rapid transformation and fusion between industry and communications systems have led to significant highway renovations, especially in self-driving. This development has affected many applications, such as the roll-out of 5G networks, the Internet of Vehicles, and the adoption of cellular vehicle-to-everything (C-V2X) connectivity. As a result, when 5G is connected, vehicles exchange traffic data, highway data, traffic signals information, roundabout information, etc., without human interference [36]. The 5G network-connected vehicles will generate a huge amount of data and have more autonomous functions. The 5G network integrated with DT can address key variables, such as capacity, reliability, mobility, latency, and security. Recently, a prediction method for 5G-enabled IoV in real-time traffic using the DT concept was introduced in [37]. Hu et al. worked in IoV solutions and introduced a DT-assisted real-time traffic data prediction model using 5G communication. As a result of this work, the proposed model proved to optimize the scheduling of traffic resources and mitigate possible traffic jams at peak times. The authors believe that the proposed method can be more accurate than others by analyzing the traffic flow and velocity data measured by IoV sensors and transmitted over 5G communications.

Deng et al. [38] proposed a combined approach consisting of DT, reinforcement learning, and expert knowledge for the self-optimization of current 5G network performances, and they described potential application scenarios for 6G to solve the problem of end-to-end delay and reduce the processing time at the local servers in many emerging critical applications. However, Dong et al. [39] adopted a digital twin framework of the current network. The proposed framework based on the DL algorithm achieved lower energy consumption with minimal computing complexity. Jagannath et al. [40] proposed an innovative DT framework as an expandable approach for data-oriented modeling and the real-time simulation of extensive systems on 5G-supported IoT networks. The DT framework employs a tiered architecture for decentralized deployment on cloud computing platforms. Additionally, it facilitates the application of AI/ML engines for event identification and prediction models. It can be concluded that the use of 5G in combination with hDT technology is essential in real-time scenarios, such as the IoV framework in which the safety and efficiency of the vehicle fleet traveling depends on the velocity and quality of the exchanged data.

### 3.5. Digital Twin and Artificial Intelligence for Autonomous Vehicles

The recent research on EVs focuses on AVs, also known as self-driving or driverless cars, i.e., vehicles driven without human intervention. Such vehicles are electric since electric propulsion is easier to be autonomously governed. With advanced technology, the vehicle will sense the surrounding environment, plan the route, and drive safely, thanks to AI and ML technology [41]. The AV is still under testing and has not yet become popular globally, but in the coming years, and due to its great benefits, the AV will occupy the global market and vehicle industry. Although the AV has been an active area of research and development in the last few decades, it still faces many challenges in developing a fully safe automated vehicle system. Considering road conditions, traffic conditions, weather conditions, and communication expansion have helped in the growth of the AV system.

Car navigation systems assist in controlling and making decisions based on prior knowledge (sensors or road map) that feeds into the system. Lopes et al. [42] proposed an efficient approach for vehicle navigation systems based on the velocity optimization paradigm. The maximum speed is adjusted to the curve of the road, and the car fol-

lows a smooth path to the lane's center. The approach is integrated into car navigation architecture and was evaluated in two separate simulators before being tested in AV prototypes. The local route and road geometry are required for autonomous driving. Therefore, Jo et al. [43] proposed a hybrid local route generation method. According to the history of the performance and the map availability, the algorithm can precisely choose the best route between the available options. The proposed method was validated and verified in real traffic conditions in an urban area in Korea. In fact, verification and validation are great challenges in AVs for safety assessment. The authors in [44] introduced a systematic review to investigate the current verification and validation software used in AVs. They discussed the simulation environments and more specific approaches, such as mutation testing, fault injection, techniques for cyber-physical systems, adversarial examples, corner cases, and formal methods.

Yang et al. [45] developed a framework that integrated the intelligent driving model with human factors, such as driving modes and their reactions and expectations on the road, to enhance autonomous driving performances. The proposed model helps to reinforce the efficiency and safety of AVs. Reinforcement learning (RL) is a widely used ML technique to train the agent on reward and punishment approaches. This approach effectively works with the AV industry as the RL algorithm learns from the driver's actions to increase a certain reward or take a decision. Masmoudi et al. [46] designed a framework for car-following based on video frame processing using RL algorithms. The framework is based on navigation decisions and automated object detection. The proposed model has achieved promising results and acceptable car-following behavior in AVs.

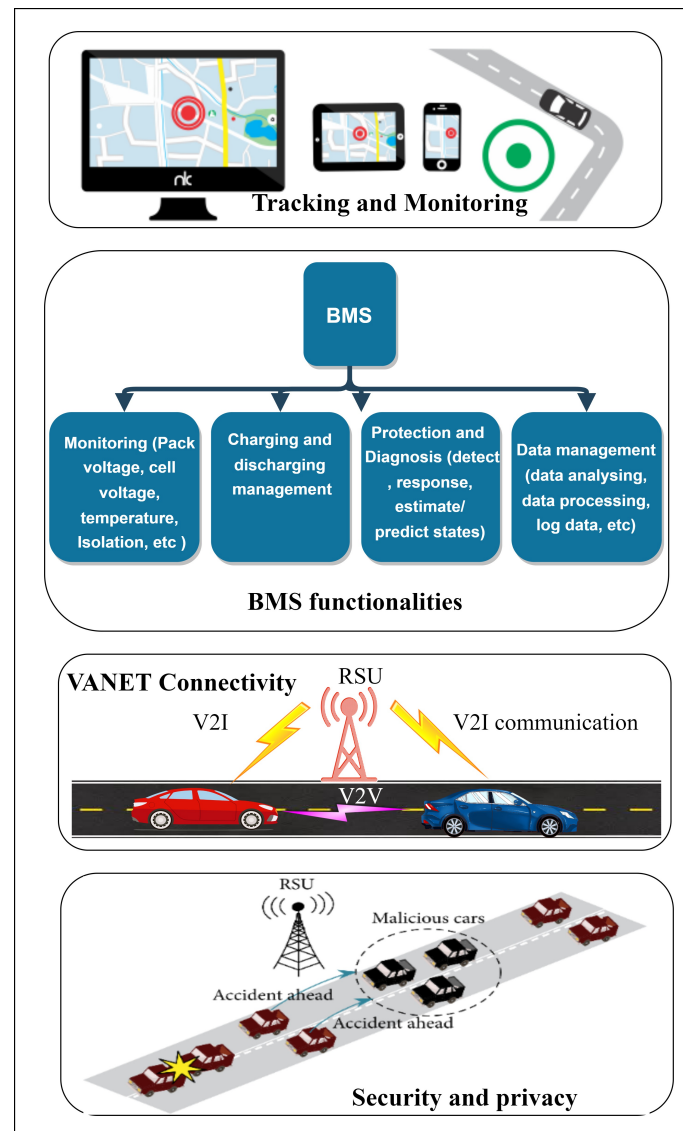
Employing RL in the AV industry is innovative and will lead to some benefits. The more information the algorithm processes, the more efficient the algorithm becomes, and the best results can be obtained. Software providers that support the DT concept have begun integrating reinforcement learning into their tools. For example, FlexSim software [47] recently introduced the RL model and the possibility of connecting external systems with the FlexSim models. Such additions will allow the software to include a 3D design tool and a data analysis tool, enhancing the concept by Rassolkin et al. [3] to specify tasks required for a specialized unsupervised prognosis and control platform for the energy system performance estimation of the AV. They developed several test platforms using DT and ML algorithms to autonomously optimize electric propulsion drive systems of self-driving electric vehicles and monitoring sensors. In addition, Venkatesan et al. [48] proposed a pre-estimation of the service requirements of EV motors for AV using intelligent DT that employs artificial neural networks and fuzzy logic in MATLAB/Simulink for the monitoring and prognosis of the permanent magnet synchronous motor distance.

### 3.6. Discussion

The performed analysis highlights DT as a promising technology for ITS, and the current literature does not exhaustively present the deployment of DT, in combination with the other discussed technologies, in EVs or AVs. Although some articles generally talk about DT technology and EVs, there is no comprehensive coverage of all of the technical aspects of applying this technology to electromobility. There is a lack of coverage on critical technologies in DT for electromobility, particularly for the next generation of self-driving systems. For instance, the use of data analytics in this area is promising as the data are crucial in the era of AI, which leads to optimizing their performance and enhancing security. During our research, we found insufficient reviews that considered the importance of data generated by ITS and how the data could be utilized in DT for the generation of value-added electromobility services; we aimed to cover this gap by providing a wider spectrum of analysis. It can be concluded that having DTs of EVs and AVs, in combination with other technological solutions, such as IoV, V2X, IoT, and 5G can facilitate technological advancements and provide the ability to optimize the single-vehicle technology and the traveling and charging operations of EV fleets.

#### 4. Current Issues in Electric Vehicles

EV development is considered a successful and promising solution to electrifying the transportation sector, and the use of EVs can lead to several environmental benefits, such as reducing noxious emissions. However, EVs are integrated with all emerging technologies, such as smart grids. This integration will result in several technical and logistic issues that could affect EV diffusion. Figure 6 shows the current and major issues affecting the electromobility sector, regarding EV tracking and monitoring, BMS, connectivity, security, and privacy issues.



**Figure 6.** Current issues in electric vehicles.

##### 4.1. Tracking and Monitoring

The use of EVs leads to important changes in mobility. Numerous advantages are linked to using EVs, including zero emissions, improved fuel economy, and lower running costs, making them perfect cars for everyday use. Security concerns, route planning, and battery charging are all challenges that vehicle monitoring systems may help to address. When planning long trips, optimal tracks, battery life, state of health, and charging time have to be planned and monitored.

An increasing number of charging stations are appearing. Since they are not as widely spread and prominent as gas stations, regular route planning and daily charging station

check-ups are necessary. The issue of range anxiety is still critical in EVs and needs to be considered in the upcoming research. Sarrafan et al. [49] proposed a novel framework to solve this issue by introducing a real-time mixed SoC estimation algorithm. The proposed system was implemented in an advanced driver assistance system. The proposed modified method includes the recalibration technique for estimating the battery state of health (SoH), the initial battery SoC, and the effective battery capacity, while considering external parameters and environmental factors, such as traveling, vehicles, traffic congestion, and driver behaviors, to make the model more accurate than the traditional models available in the literature where the environmental conditions and driver behaviors are not considered. Both laboratory and field tests have been conducted using a Nissan Leaf and, as a result, the SoC estimation of lithium-ion batteries improved with great accuracy and high real-time capability. In [50], a discrete scheduling process was formulated to track arbitrary power profiles and control charging, limiting it to the maximum rated power. The coordinated charging of PEVs is formalized by considering the realistic case of PEVs with different charging rates. In addition, a novel algorithm has been developed to ensure plug-in EV charging without the need for a central aggregator. It ensures that the power profile imposed by the power utility is tracked and not exceeded while maximizing user comfort. The effectiveness of the algorithm has been demonstrated in realistic scenarios, with a heterogeneous PEV population, but no performance comparison with other methods has been provided.

Thus, vehicle monitoring systems greatly affect research and development and improve the operations of the industrialization of EVs. In addition, Wang et al. [51] designed a remote monitoring system for EVs, i.e., a combination of remote monitoring platforms and onboard terminals installed on the vehicle. On the contrary, with respect to [50], the idea was to aggregate the data from the onboard terminal, such as the battery status, temperature, SoC, running status, etc., and send them to the server cluster model for processing and monitoring. This model can improve communication reliability and enhance the real-time performance of the system. The system structure of the electric vehicle remote monitoring system is based on CAN and GPRS technology and aims to provide a certain guiding significance for the design of remote real-time monitoring systems for new energy vehicles. It is necessary to monitor the parameters of EVs to avoid electrical malfunctions because problems of this type are costly to solve. In this context, adopting the DT concept can provide solutions to the discussed issues about EV monitoring and tracking.

#### 4.2. Battery Management Systems

Worldwide, there are serious challenges, such as global warming and greenhouse gas emissions due to the use of petrol and diesel in vehicles, which result in excessive levels of (CO<sub>2</sub>). EVs are now being promoted and widely regarded as eco-friendly and substitutes for other vehicles based on combustion engines. Rechargeable batteries are widely used as the power supply for electric vehicles. There are many different types of batteries, including lithium-ion, lead-acid, nickel-cadmium, and nickel-metal hydride [52]. The lithium-ion battery is the most used battery, which provides a high density of electric energy, eco-sustainability, and a long life cycle. To keep the EV battery's state of health at a good quality level, we must take proper care of the battery and adhere to proper charging operations (over-charging, current, voltage, or discharging). These operations may cause problems and damage the battery and the EV.

Several studies have been conducted on applying IoT and cloud computing to solve BMS issues. Recently, Kim et al. [53] developed a cloud-based battery system to monitor stationary batteries and help in fault diagnosis platforms at a large scale. As the authors claimed, it is a new model, with no comparison made in the previous literature. It incorporates IoT-enabled wireless battery module management systems and a proposed cloud battery management platform to support onboard battery health monitoring and provide intelligent, cost-effective maintenance for large-scale battery energy storage systems (BESSs). The hybrid filter (HF)-based condition monitoring algorithm of cells and



the proposed outlier detection-based fault diagnosis algorithm are implemented in the platform. Compared to the Kalman filter family and sliding mode observer, the HF leads to less computational costs and chattering issues, respectively.

Tanizawa et al. [54] introduced a cloud framework for EVs to manage the battery information related to the battery replacement system. This cloud-connected battery management system will maximize the value of the shared batteries by using location data cloud to continuously connect to the batteries, manage the SoC, and monitor changes in their characteristics. Friansa et al. [55] presented a solution for battery monitoring in a microgrid system based on IoT, but different from [53], the authors presented a smart microgrid by integrating a battery pack, PV system, intelligent electronic device (IED) hybrid inverter, grid connection, and electricity load, where no fault diagnosis was considered. In this framework, the IoT is realized through a communication channel with the IED, data acquisition algorithm, cloud system, and human-machine interface (HMI). The data stored in the cloud system database are processed and analyzed to produce information that can be accessed by users through desktop and mobile devices using the ExtJS/HTML5 framework. The analytical results show good performances for the average execution and connection times within the architecture for the overall BMS-IoT-based data acquisition to the cloud server. Moreover, the availability of monitored data shows satisfactory results for the reliability of the BMS-IoT system data acquisition.

The above studies have some drawbacks as the technical details were not introduced in the cloud and the battery diagnostic algorithms that can help improve accuracy and data storage in the cloud were not exhaustively analyzed. Therefore, Li et al. [56] attempted to overcome these drawbacks by introducing a cloud BMS for battery systems to improve the computational power and data storage capability of cloud computing. The proposed cloud computing system provides computational power and data storage capacity with greater speed and utility. In EVs, charging operation is done by a graded control structure controlled by an aggregator, which controls all EV charging rates. For example, Nour et al. [57] proposed a new approach for smart charging in EVs. A fuzzy logic controller was used to control and manage the EV charging process to maximize the electric utility and EV owner benefits. In particular, the controller regulates the EV charging power based on the electricity price signal provided by the electric utility and EV battery SoC. The proposed technique was evaluated using a simulation with MATLAB/Simulink, and the EV charging impact on the distribution network decreased compared with uncontrolled charging. Many EV drivers still face a limited driving range. However, the range of most EVs has considerably improved in only a few years by increasing the battery size and improving lithium-ion battery technology. Nonetheless, the current EV range is still not convenient for users. The range of an electric vehicle and its higher purchase cost are two of the main deterrents to the widespread use of EVs [58]. Therefore, we need to work around this limitation of the driving range in the future and the DT can support the further necessary advancements of battery technology.

#### 4.3. Connectivity

Connected cars will inevitably improve user experiences and the ability to manage the network of interconnected vehicles. Two types of connectivity can be distinguished: intra-vehicle connectivity and inter-vehicle connectivity. The bandwidth requirements in vehicles have increased dramatically because of recent innovations and changes in automotive technology. However, modern vehicles have gradually developed in terms of entertainment and networking with advanced capabilities [59]. Intra-vehicle and/or internal networks are designed to share data across the various sub-systems, ECUs, sensors, and actuators, so that a single vehicle can operate easily. Although the sensors and networks are exclusive to original vehicle companies, these technologies typically meet standards that permit vehicle diagnostics and future applications to communicate with the vehicle using external technologies or devices.

Inter-vehicle connectivity is a networking approach that allows data to be transferred from the vehicle to other vehicles, remote computers, and other cloud infrastructures. Remote applications can combine the vehicle's data with external sources (such as traffic or weather data) [60]. Gupta and Sandhu [61] developed an authorization framework by describing data exchange scenarios on the Internet of Vehicles (IoV). The development of IoV has produced a considerable amount of real-time traffic data. These traffic data are used to generate a kind of digital twin that connects the physical vehicles and their virtual representations via 5G. Moreover, Khan et al. [62] presented an effective communication framework to mitigate cyber attacks on connected and autonomous vehicles (CAVs).

In this context, data will be collected from different networks and heterogeneous resources, including charge management software, weather sites, live EV modeling, battery modeling, telemetry devices, and route maps, such as Google Maps. Data integration in such platforms is a difficult task and needs to support data interoperability in connected vehicles to communicate effectively and efficiently. Reference [63] deals with AV collaboration as a service and proposes a DT-based scheme to facilitate collaborative and distributed autonomous driving. Specifically, a DT is designed for each AV, and a DT-enabled architecture is developed to help AVs make collaborative driving decisions in virtual networks. With this architecture, an auction game-based collaborative driving mechanism is then used to decide each group's head DT and tail DT. After that, by considering each group's computation cost and transmission cost, a coalition game-based distributed driving mechanism is developed to decide the optimal group distribution for minimizing the driving cost of each DT.

#### 4.4. Security and Privacy

The security issues seriously influence the CAV sector due to the comprehensive connectivity over the internet and cloud. The essential threat in CAV derives from exchanging information with other vehicles and servers. The CAV can share the ID details, battery information, or vehicle location. The data exchange between CAVs and the server might be subject to cyber attacks, such as (DoS, spoofing, privacy attacks, modification, etc.). Several studies in the literature surveyed the issues of security and privacy in electromobility. Many approaches and technologies have been innovated to increase the protection level of the EV. Guo et al. [64] proposed an anomaly detection system to enhance the cyber security of the steering stability control system in the CAV. The proposed approach can identify the threats to control inputs and sensors by combining two approaches, physics-based and learning-based approaches. The results have shown an improvement in the cyber-physical security of EVs. Babu et al. [65] introduced a robust authentication protocol for charging electric cars while driving. Secure and lightweight primitives, such as elliptic curves and hash functions, are used in the proposed protocol.

The protocol's security is being investigated to show that it can handle different attacks. Kavousi-Fard et al. [66] developed a cyber-resistive model for detecting vehicular cyber attacks. A hybrid smart structure built of wavelet decomposition technology and a modified support vector machine can detect any malicious behavior in the controller area network bus. The proposed approach performs excellently in detecting cyber attack communications while also detecting regular data. Thus, it is essential to find a way to validate the data exchange and transformation between the vehicles and infrastructure before the system implementation. The DT will give the ability to validate the security and privacy of the IoV by simulating the network scheme and using AI and ML technologies; it will allow for predicting threats that can be harmful to the vehicular network [67]. Safety and security functions are of basic importance in AV. Reference [68] aimed to identify a standard vehicular DT framework that facilitates the data collection, processing, and analytics phases. The DT was explored to automate the decision-making process inside an AV using radar sensor data collected from the initial, analytic, and reporting phases, and generate reports to be sent to AVs. The recommended model aims to identify, analyze, and assess the threats, and provide the user with an opportunity to take appropriate countermeasures in ensuring

safety and security using DTs in driverless vehicles. Some of the advantages presented using the car-follower model can reduce the risks of cyber attacks and accidents.

#### 4.5. Data Analytics

Big Data analytics influence ITSs by enabling EV internet connections to optimize their performances. Plug-in EVs should be connected and able to revolutionize energy use, creation, and redirection. Smart grids and EVs generate equal amounts of significant Big Data among the connected devices. As consumers and Big Data producers, EVs produce data from various sources, such as sensors, logs, etc. By utilizing Big Data technologies, the data can be used to develop smart charging algorithms, solve energy efficiency issues, develop policies and strategies for the location of electric charging stations, and turn smart cities into green cities [69].

Technology integration transforms the transport and automotive industries by tracking, analyzing, and evaluating the demographics of EVs. The demographic data of EVs include statistics on charging stations, battery features, the analysis of energy usage, and route profiles. In this framework, the goal is to overcome obstacles, such as battery capacity, battery prices, charge times, and the availability of charging stations to take full advantage of the EV's potential. Using data science and AI technology could improve operations and overcome the main obstacles [70]. Nowadays, every industry is affected by the role of data. We are facing an increase in the volume of data produced, particularly in transportation. The transportation industry promised to develop better information systems to optimize energy consumption in highly complex environments through the electrification of vehicles. Car manufacturers, governments, and charging infrastructure providers utilize data analysis and data science tools to use and analyze the available data to provide services for optimizing EV use.

Big Data analytics is often used to assess the driving range and effectively reduce user anxiety. In [71], the authors propose an approach for classifying the range estimate. The data are classified into standard, historical, and real-time data. In addition to the range estimation, data produced from EVs can be used to determine the position of public charging stations. Different types of information, including traffic density, gas station distribution, and vehicle ownership, are used in this respect. In [72], the authors proposed a solution to site public EV charging stations using a large-scale trajectory dataset. They examined the model based on data from 11,880 taxis in Beijing. Security is one of the most important features that data analysis offers for ITS. In this direction, various ML and DL technologies have been developed to secure the transportation systems and improve prediction accuracy. Lv et al. [16] developed a DT framework based on the DL algorithm, which was combined using the convolutional neural network and support vector regression. The proposed model can reduce the system data transmission delay, improve prediction accuracy, and reasonably modify pathways to reduce traffic congestion. By combining the power of predictive analytics and data intelligence, the predictive maintenance analysis of batteries can be improved. The integration of these technologies aims to achieve high battery efficiency and reliability. Sreedhar et al. [73] introduced a simulation design of common BMSs. The proposed system was designed to handle and control battery parameter values, such as the battery voltage, consumption of power, current, and SoC.

Data analytics have rapidly improved EV efficiency, and DT technology provides smart modeling capabilities in this field. Tang et al. [74] proposed a novel model utilizing a machine learning algorithm to increase computational efficiency and precision. In smart EVs, it is also important to focus on other essential aspects, such as the safety of drivers and passengers. A safety-based intelligent approach in EV, presented in [75], uses a fuzzy adaptive control method for vehicle following, decreasing accident rates. Another safety application proposed by Guo et al. [76] involves direct yaw control, which enables an EV to keep the vehicle within the allowed path and increase the stability of the steering system. Behrendt [67] developed a hybrid architecture using DT technology to perform analytics operations and increase real-time driver safety. The information that comes from

the physical smart vehicle through sensors is analyzed to detect privacy anomalies in the transportation ecosystem and minimize other privacy risks. As a result, data science, AI, and Big Data tools are emerging use cases in this context. They can significantly impact the current EV market to improve the end-product performance.

#### 4.6. Intrusion Detection Systems

Network security has become a critical research subject due to the advanced development of internet and communication technologies over the previous decade. It uses firewalls, antivirus software, and intrusion detection systems (IDSs) to keep the network and its assets safe in cyberspace. Therefore, researchers have conducted many studies on network security, with various approaches and technologies employed to develop algorithms for detecting unusual activities on different network platforms [77].

To meet the network security needs in electromobility, various IDS have been developed. IDS is a security framework that continuously monitors network traffic to detect any abnormal behavior that may violate the network policy and threaten its confidentiality, integrity, or availability. The vehicular IDS architecture proposed by Loukas et al. [78] ensures onboard IDS, collaborative detection, and offloaded intrusion detection; onboard IDS is where the vehicle identifies illegal behavior on networks on its own; collaborative detection is where vehicles collaborate to determine whether they are under attack or not; offloaded intrusion detection is where the detection process is conducted in the cloud. In this context, Zeng et al. [79] developed end-to-end intrusion detection using a DL algorithm to detect malware intrusions for onboard units. Different from previous intrusion detection methods, the proposed method only requires raw traffic instead of private information features extracted by humans. The performance was compared with previous methods on a public dataset and a simulated real-life VANET dataset. The experimentations show that the model can achieve higher performance with minimum resource requirements.

Shams et al. [80] introduced a trust-aware-based IDS; they used the combination of a modified promiscuous mode and support vector machine to ensure the safety of vehicles and detect malicious behavior in any network node. Liang et al. [81] proposed an intelligent IDS model based on the hidden Markov methodology to filter out malicious messages and reduce the detection time without affecting the detection rate. The authors claim that it is the first work in the literature to model the state pattern of each vehicle in VANET as a hidden Markov model (HMM) to quickly filter the messages from the vehicles instead of detecting these messages. It consists of three modules: schedule, filter, and update. In the schedule module, the Baum–Welch algorithm is used to produce a HMM and its parameters for each neighbor vehicle. In the filter module, multiple HMMs are used with their parameters to forecast the future states of neighbor vehicles in which the messages from them are filtered. In the update module, a timeliness method is used to update HMMs and their parameters. The performed experiments show that the IDS with FM-HMM performs better in terms of the detection rate, detection time, and overhead. Sedjelmaci et al. [82] designed an efficient, lightweight IDS simulator for the vehicular network. The model can protect vehicular ad hoc networks against denial-of-service attacks, false alert generation, and integrity target attacks. They used the NS-3 simulator (a discrete event network simulator for Internet systems) to present the detection mechanism's performance analysis. The simulation framework shows high-level security and an accurate detection rate.

#### 4.7. Discussion

In this section, we will discuss the main challenges for electromobility and how DT technology is used and can be used in future applications to address them. The complexity of integrating ITS with emerging technologies and communication tools using the DT concept is also emphasized. In this part, we present several technical and logistic issues under different topics that could represent risks for the future of smart and green mobility. Several works have been conducted by researchers to find effective solutions, especially for

monitoring and tracking vehicles, replicating battery and vehicle technology to improve energy consumption, charging times, driving ranges, etc.

In some cases, DTs of batteries and vehicles have helped to perform simulations and validate experiments, leading to better performances and pushing forward technological advancements. Future works must further develop technologies and tools that are able to answer to the needs of connected and smart EV fleets in which large amounts of data need to be safely exchanged in real time, and collected and elaborated on, while respecting security and privacy issues. In the next section, taking into account the previous challenges and issues, we will discuss the benefits and significant challenges in EV networks and how the DT could provide solutions that help integrate modern technologies and provide the best services at lower costs.

## 5. DT Solutions and Future Directions

The DT concept involves feeding data from the real world back into the virtual environment to improve model accuracy. This approach reduces the gap between the real and virtual worlds, allowing for real-world simulation. As a result of this paper, we present the important challenges in EV networks and how the DT can provide solutions.

### 5.1. Cost-Effective and Reliability

The implementation of EV networks is one of the most relevant challenges due to the shortage of infrastructure and safety measurements. Providing infrastructure services to implement EV networks is very cost-effective. The maintenance of EV services is also cost-effective. The DT model of the EV can be evaluated before deployment, thus lowering maintenance costs and making DT a cost-effective option. The present state of the EV system does not have reliability in data transmission. Reliability is the main challenge in EV transportation systems to operate EVs safely under various conditions. In the future, EV transportation systems should maintain data reliability and scalability.

### 5.2. Visualization and Charging Time

The data visualization provides a complete overview for EV consumers to plan long-distance transportation. However, the EV system has limitations in visualization, which can lead to problems in testing EV functions and efficiency. The DT integrates 3D graphics and audio with real-world objects to solve this issue by using IoT and AI. Using such technologies, the operator can monitor and control the EVs and allow them to communicate and interact with the DT model to improve efficiency during and after the design process. Whether the charging system is standard, fast, or quick, the charging time is still quite long. This is one of the main reasons that is holding back the growth of the EV industry. There is also a need to conduct research on wireless charging. The DT simulation can improve the charging time by analyzing the data from the virtual model, and the result can be used to evaluate charging infrastructures and charging efficiency.

### 5.3. Intrusion Detection Systems for EV and AV

Data attacks are greatly mitigated with DT technology, which provides greater security for the mechanism against attacks and protects the privacy of EV users. The DT enables the development of high-accuracy models for real-time systems using massive quantities of operational data with expert observations. The DT architecture could be built to effectively detect intrusions or anomalies that behave abnormally inside the vehicular (EV and AV) network. Available studies have explored this technology for industrial applications using data analytics and ML technologies, such as the one-class support vector machine, local outlier factor, and deep unsupervised learning. For example, in [83], Fraser et al. proposed DT architectural enhancements to improve the security of unmanned aerial systems. Gao et al. [84] introduced an anomaly detection system to monitor abnormal behaviors in DT-based cyber-physical systems.



Gehrmann et al. [85] investigated how the DT model and security architecture can share data and control critical security processes. They introduced a new security framework that will provide the foundation for future research work in automation and control systems. Xu et al. [86] presented a two-phase digital-twin-assisted fault approach based on deep transfer learning; it detects faults during the development and maintenance of a vehicle's body-side production line. Snijders et al. [87] used the convolutional neural network model to improve the predictive power of the DT for cyber-physical energy systems. The behaviors of ten batteries were predicted using real-world data. They concluded that ML for DT can aid in maintaining a heterogeneous energy ecosystem. Castellani et al. [88] presented cluster centers (a clustering-based method) and Siamese autoencoders (a neural architecture designed for weakly supervised environments with few labeled data samples). The methods utilize a DT model to create a training dataset that replicates the machinery's usual operation. The above research employed DT to detect intrusions and abnormal behaviors in unmanned aerial systems, manufacturing, and energy systems utilizing ML and DL. In conclusion, the same technologies could be applied to EVs and AVs to obtain benefits.

#### 5.4. Case Studies

Reference [89] establishes and compares a traffic infrastructure efficiency assessment model using data envelopment analysis based on DT and a traffic flow prediction model based on long short-term memory. The traffic flow data of a certain road section in Zhenjiang City were simulated and predicted. Using the transportation infrastructure of 12 cities in J province as the research object, the two models were verified to maximize the potential of intelligent transportation facilities. The results show that the established DEA model based on DT can estimate the efficiency of transportation infrastructure more reasonably and accurately. Compared with other models, the traffic flow prediction model based on long short-term memory is more accurate in traffic flow prediction, which can provide a reference for intelligent transportation system infrastructure investment planning.

A case study on safety and security functions in autonomous vehicles was tested in [63] to demonstrate the effectiveness of the proposed approach. A vehicle follower model was analyzed when radar sensor measurements were manipulated to cause a collision. Almeida et al. [68] proposed a standard framework for digital twins that facilitates data collection from vehicular networks, processing, and analytics. A vehicle follower case study was proposed to prove the model's efficiency. The vehicular model was analyzed during the manipulation of measurements of the radar sensors in an attempt to make a collision. This research can guide the direction of future research using the DT concept in the EV and AV industries. Another case study proposed by [61] introduced a real-time case study on smart cities to improve the defense mechanisms for connected AV attacks. To reduce security and privacy attacks, such as DOS, hijacking, man-in-the-middle, GPS spoofing, privacy attacks, and replay attacks, the authors proposed a blockchain-based architecture that provides a secure and decentralized connected AV.

#### 6. Review Summary

Table 1 shows that there are interesting works in DT from different perspectives and applications. The table classifies the existing contributions in this field based on the use of DT in combination with other technologies and services.

This literature is based on the classification of the main issues and challenges of electric and autonomous mobility, exploiting the DT technology and tools in combination with modern technologies related to the automation, telecommunications, and ICT fields as shown in Table 2. The proposed review can be a reference for researchers and practitioners who want to have an overview of the (current and future) use of DT for supporting the technological growth of EVs and AVs.

**Table 1.** Paper classification based on the technologies and services applied for DT.

| Services Technology | BMS and Charging    | User Experience | Tracking Monitoring and Control | Privacy and Security | Reviews                                                                                                                         |
|---------------------|---------------------|-----------------|---------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 5G & Networks       | [52]                | [31,32]         | [25,33,35]                      | [59,60,75,76]        | [5–7,9,14] AI, ML, and Big Data in Digital Twinning<br>[8] DT for cyber physical system                                         |
| AI & ML             | [19,35,65,67,68]    | [16,17,20]      | [15,37,64,69]                   | [61,66,74,79]        | [22] Survey on Digital Twin Networks<br>[10,24] Enabling technologies, challenges and open research in manufacturing using IIoT |
| IoT and cloud       | [20,29,30,45,48–51] | [18,21,55]      | [12,13,23]                      | [11,56]              | [39] A Systematic Literature Review                                                                                             |
| VS and IoV          | [26,27,44]          | [2,28]          | [46,70]                         | [58,77]              | [53,79] A review of electric vehicle life-cycle emissions                                                                       |
| AV                  | -                   | [40]            | [3,37,38,41,43]                 | [57,71,78,80]        | [72] review of current ML approaches for anomaly detection                                                                      |

**Table 2.** Digital twin review work comparison based on the analyzed technologies.

|                         | Big Data & ML | IoT | Digital Digital Manufacturing | Security | 5G  | Mobility | AV  |
|-------------------------|---------------|-----|-------------------------------|----------|-----|----------|-----|
| Q.L et al. [5]          | -             | Yes | Yes                           | -        | -   | -        | -   |
| Grieves [6]             | -             | Yes | Yes                           | -        | -   | -        | -   |
| Weyer et al. [8] Yes    | Yes           | Yes | -                             | -        | Yes | -        | -   |
| Rathore et al. [13]     | Yes           | Yes | Yes                           | Yes      | -   | Yes      | -   |
| Bhatti et al. [19]      | Yes           | Yes | Yes                           | -        | -   | Yes      | -   |
| Kharachenko et al. [14] | Yes           | -   | Yes                           | Yes      | -   | Yes      | -   |
| Tao et al. [16]         | Yes           | -   | Yes                           | -        | -   | Yes      | -   |
| Fuller et al. [27]      | Yes           | Yes | Yes                           | -        | Yes | Yes      | -   |
| Rajabli et al. [42]     | Yes           | Yes | -                             | Yes      | -   | Yes      | Yes |
| Wu et al. [47]          | Yes           | Yes | -                             | -        | Yes | Yes      | -   |
| Vidhi et al. [57]       | -             | -   | Yes                           | -        | -   | Yes      | -   |
| Ali et al. [76]         | Yes           | Yes | -                             | Yes      | Yes | -        | -   |
| Our Survey              | Yes           | Yes | Yes                           | Yes      | Yes | Yes      | Yes |

## 7. Conclusions

This paper provides a complete survey of the existing literature in the last five years on the use of DT technology in Intelligent Transportation Systems. The awareness of DT has been growing exponentially due to the number of applications that demonstrate its capabilities for connecting the physical and digital worlds. EVs are physical objects for which DT can overcome some important limitations and enhance their functionalities. There is a lack of coverage on the analysis of critical technologies in DT for electromobility, especially for the next generation of self-driving systems. We found insufficient reviews that consider the importance of data generated by ITS and how it could be utilized in DT to allow for the creation of value-added services.

The paper addresses the complexity of integrating ITS with emerging technologies and communication tools using the DT concept to accelerate the technological advancement goals and achievements of EVs and AVs. It highlights the potential of DT technology based on several aspects, such as tracking and monitoring, security and privacy, data analytics, and intrusion detection, which could enhance efficient electric and autonomous vehicle network management.

Since the DT application is relatively new and rapidly changing, the presented review only focuses on articles published in the last five years. A future analysis will use AI techniques to perform a larger and more accurate review.

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## Abbreviations

The following abbreviations are used in this manuscript:

| Abbreviation | Description                       |
|--------------|-----------------------------------|
| DT           | Digital Twin                      |
| DTN          | Digital Twin Networks             |
| IoT          | Internet of Things                |
| AI           | Artificial Intelligence           |
| ML           | Machine Learning                  |
| EV           | Electric Vehicle                  |
| VS           | Virtual Sensors                   |
| SoC          | State of Charge                   |
| VR           | Virtual Reality                   |
| AR           | Augmented Reality                 |
| ITS          | Intelligent Transportation System |
| AV           | Autonomous Vehicle                |
| ADAS         | Advanced Driver Assistance System |
| CPS          | Cyber-Physical System             |
| ECU          | Electronic Control Unit           |
| BMS          | Battery Management System         |
| V2X          | Vehicle-to-Everything             |
| RL           | Reinforcement Learning            |
| IDS          | Intrusion Detection System        |

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